IoT Authentication Protocols: Classification, Trend and Opportunities

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(Survey-Tutorial Paper)

***Abstract*—This paper reviews three main aspects of authentication protocols of Internet of Things (IoT): classifications and limitations, current trends, and opportunities. First, we explore the significance of IoT authentication protocols in ensuring secure communication and the protection of transmitted and received data, focusing on the classifications and associated limitations. Second, we discuss the latest developments and trends, such as using blockchain technology and machine learning to enhance authentication protocols. Third, we highlight the future opportunities, including the development of human-centric authentication designs and improved platform interoperability. At the end of this paper, we provided some insights gained for the new researcher, offering analyses of the trends and challenges in this field, giving recommendations for improving IoT authentication protocols, and emphasizing the need for further research and cooperation to develop advanced security solutions.**

***Index Terms*—Authentication protocols, biometric authentication, blockchain technology in IoT, Internet of Things (IoT), IoT authentication, IoT device security, machine learning in authentication, smart home security.**

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I. Introduction

In TODAY'S world, marked by rapid technological progress and complex tech systems, the Internet of Things (IoT) is an important foundation of this advancement. This incredible technology enables the connection of a wide range of devices to the internet, from small and mobile smart devices to large and complex industrial systems, thereby creating new possibilities for the use of information technology (IT) and communication applications in all spheres of life [1].

As we rely more on smart devices and interconnected systems, protecting these devices and protecting the data exchanged between them becomes very important. Here, authentication plays a pivotal role in securing IoT networks. It is the technical process that verifies the identity of the devices and the validity of the data exchanged between them. It works as a protective barrier to protect these systems from potential cyber attacks or unauthorized manipulation [2].

The importance of authentication in the IoT is highlighted by its ability to provide an extra layer of security, protecting devices and data from cyberattacks and information leaks. Authentication verifies the legitimacy of connected devices and ensures that the data exchanged are safe and secure [3]. It also enhances trust between interconnected devices and contributes to the efficiency and performance of systems, since the demand for security and privacy is on the rise, it is important to keep up with the growing requirements [4].

This motivated us to provide more details on authentication protocols classifications, including password-based authentication, digital certificate authentication, biometric authentication, two-factor authentication, and multi-factor authentication. This study aims to address the existing gaps in the literature by not only classifying traditional authentication protocols in IoT but also by pioneering the exploring the latest trends in advanced technologies such as blockchain and machine learning. Unlike previous surveys, our research focuses on developing sustainable and secure authentication methods that enhance cross-platform interoperability and user-centered design, setting the stage for future-proof solutions in IoT security. Each class offers specific advantages related to the level of security and efficiency in handling data security challenges [5], [6]. Blockchain and machine learning as key elements in the development of authentication mechanisms [7] are trends and attracted many researchers. Blockchain technology offers unique features in securing transaction records and enhancing transparency, while machine learning helps with data analysis and the precise prediction of potential threats. An example of applications in this field includes using blockchain to protect medical data in healthcare systems and employing machine learning to detect early cyberattacks in industrial systems [8], [9].

As technology advances, so do the solutions and strategies to support IoT security. From blockchain robustness in transaction security to machine learning precision in threat detection, the possibilities are endless. These advancements are not just technical feats, representing a significant leap toward a more secure and reliable digital future [10]. In this review, we present a view of the challenges and opportunities in IoT security, focusing on the role of authentication in securing these networks and contributing to their sustainable and secure development. By highlighting the latest developments and trends in this field, the research aims to guide researchers and developers toward new innovations that improve and enhance IoT security.

A. Motivation

The rapid advancement and increasing adoption of IoT technologies make robust and reliable authentication methods critical; IoT devices are frequently targeted by various security threats and vulnerabilities, which require innovative authentication protocols to protect data integrity and confidentiality. This study addresses these challenges by exploring different classifications and authentication methods and analyzing their effectiveness in improving security and trust in IoT environments [11].

With all these motivations, this paper answers three research questions:

- How do different classifications and methods of authentication contribute to enhancing security and trust in the world of connected things (IoT), and what are the challenges and limitations they face? (This question aligns with the content discussed in Section II)
- How do advanced technologies like blockchain and machine learning contribute to enhancing the security of smart devices and evolving authentication systems in IoT? (This question is answered in Section III).
- How can a standardized authentication framework and human-centered design contribute to enhancing system integration and improving user experience in the world of IoT? (This aligns and corresponds to the detailed content discussed in Section IV).

B. Contributions

The primary contributions are as follows:

1) *Classifications and Limitations:* We classify the authentication protocols, including passwords, certificates, biometrics, two-factor, and multi-factor, and explore the advantages, disadvantages, and limitations of each class and show how they enhance security in IoT environments. We also provide detailed information on the technical aspects of each class and explain why it is crucial to protect devices and data.
2) *Trends:* We explore the latest trends in IoT authentication protocols, such as blockchain and machine learning technologies. We also discuss integrating identities to enhance communication security between devices and offer valuable insights that could contribute to the future of security and privacy in IoT authentication.
3) *Opportunities:* We review the new opportunities in identity verification protocols of IoT, covering the utilization of standardized methods to authenticate identities and secure data exchanged by devices. Then, we discussed the cross-platform interoperability and confirm the importance of designing solutions that focus on human needs. Furthermore, we explore how technologies like face and fingerprint recognition can act as a balance between being safe and easy to use while keeping an eye on being exact and respecting privacy.

C. Related Surveys

The related surveys are listed in Table I, emphasizing various aspects of IoT authentication protocols, highlighting identified gaps, technologies used, applications covered, and methodologies. Identifies issues such as security challenges, limitations of biometric methods, and vulnerabilities in current protocols, demonstrating the need for a new survey. This new survey integrates recent advances such as advanced encryption, key management, PUFs, and blockchain integration to enhance security and efficiency in different IoT environments. By addressing these gaps and exploiting new opportunities, the survey aims to improve the overall security and effectiveness of IoT authentication systems.

When reviewing previous studies on authentication protocols in the IoT, it becomes clear that most research focuses on classifying traditional protocols, such as password-based and digital certificate-based authentication, while sometimes highlighting biometric and multi-factor authentication. While these studies provide a comprehensive overview of current challenges and limitations, they often avoid delving into the detailed integration of modern technologies like machine learning and blockchain to enhance security.

Furthermore, our research offers a unique contribution by classifying authentication protocols and examining how they enhance security in IoT environments. We explain the importance of protecting devices and data, explore the latest trends in IoT authentication protocols, and discuss the integration of identities to improve communication security between devices. This study provides valuable information that could contribute to the future of security and privacy in IoT authentication. Also, we explore blockchain and machine learning technologies to improve existing authentication protocols and explore new applications. We specifically focus on developing authentication protocols that consider both sustainability and security, incorporating advanced technologies to ensure greater data protection and greater compatibility between different systems.

Additionally, we explore new opportunities, such as developing user-oriented authentication designs that prioritize user experience and ease of use, as well as improving interoperability

TABLE I
A COMPERHENSIVE OVERVIEW OF CUTTING-EDGE AUTHENTICATION RESEARCH

Reference (Year)	Gaps Identified	Technologies Used	Applications Covered	Methodology
[5] (2023)	Highlights challenges in secure authentication and key management in IoT environments, particularly issues like data tampering, unauthorized access, and security threats in digital security and privacy	Explores various technologies, including cryptographic algorithms, key management schemes, and authentication mechanisms, focusing on lightweight and resource-efficient solutions for IoT devices	Wide range of IoT environments such as healthcare, smart farming, intelligent transportation, and smart cities, focusing on secure and efficient data communication and management	Systematic literature review, classification of existing authentication and key management schemes, analysis of their security and performance, and identification of challenges and future research directions
[11] (2023)	Identifies gaps in secure authentication and key management within IoT environments, particularly focusing on issues such as data tampering, unauthorized access, and general security threats	Explores various technologies including cryptographic algorithms, key management schemes, and authentication mechanisms, focusing on lightweight and resource-efficient solutions	Healthcare, smart farming, intelligent transportation, and smart cities, focusing on secure and efficient data communication and management	Systematic literature review, classification of existing schemes, security and performance analysis, and identification of challenges and future research directions
[12] (2023)	Highlights limitations in current biometric methods, such as computational expense and practical challenges in implementing advanced techniques	Discusses gait recognition (2D and 3D), fingerprint-based methods, hybrid systems, sensor-based and video-based methods	Surveillance systems, mobile device authentication, and general biometric security implementations	Literature review, empirical performance analysis, and proposing improved methods to address identified gaps
[1] (2022)	Hardware and software constraints in IoT environments, increased cyberattacks on IoT devices, and a lack of advanced protection strategies	Text-based methods such as passwords and PINs, biometric methods like fingerprints and facial recognition, web-based protocols, and hardware tokens and keys	Smart devices and IoT systems, such as smart cameras, smart homes, and building automation, as well as general and specific IoT applications including access control, smart transportation, and medical devices	Comprehensive literature review, empirical performance and vulnerability analysis, computational complexity analysis, and the proposal of mitigation strategies
[2] (2021)	Dynamic data flow causing network paralysis, challenges in ensuring valid user access in WSNs, and problems with heterogeneity, scalability, and trustworthiness in edge computing	Use protocol Elliptic Curve Cryptography (ECC), chebyshev polynomial, hashing functions, session passwords, and encryption for multi-level authentication	Secure communication in IoT, focusing on smart homes, vehicular networks, and industrial IoT	Literature review, development of a multi-level authentication protocol, and performance evaluation through simulations of blackhole and DoS attacks, assessing detection rate, PDR, and QoS
[13] (2022)	An identifies traditional IoT vulnerabilities (device intrusion, protocol weaknesses) and new challenges from the TAP model, such as inconsistent device states and lack of cross-platform authentication	Defense technologies include software upgrades, firmware encryption, machine learning-based anomaly detection, device behavior recognition, traffic filtering, and rule-based automation	Focuses on smart home services like security monitoring, device control, and automation of lighting and climate control	Involves literature review, classification of attacks and defenses, analysis of security requirements, and identification of future research directions
[14] (2021)	An identifies gaps such as the risk of single-point-of-failure in password-based single-sign-on systems and vulnerabilities to dictionary guessing attacks (DGA) and online password testing attacks (OPTA)	Threshold cryptography, password hashing, and server-side key sharing to enhance security in single-sign-on authentication schemes	Focuses on mobile environments, particularly secure access to various mobile services through single-sign-on authentication, such as mobile payment systems and service providers' applications	Literature review, the development of the PROTECT authentication scheme, and performance evaluation through simulations to test efficiency and resistance to various attacks.
[15] (2022)	The vulnerabilities in PUF-based AKA protocols to modeling attacks, and challenges in achieving scalability and robustness in IoT, WSNs, and smart grids	Uses PUFs, CRPs, and cryptographic techniques including ECC and symmetric cryptography	The applications include IoT, WSNs, and smart grids, focusing on secure communication and authentication in these environments	Comprehensive literature review, classification of existing PUF-based AKA protocols, analysis of their security and performance, and identification of challenges and potential improvements
[16] (2022)	Identifies the need for protocols that ensure security while being computationally efficient and resistant to physical attacks, such as side-channel and cloning attacks	Uses PUFs combined with cryptographic techniques to create secure authentication and key exchange protocols	Various IoT devices in peer-to-peer networks, with a focus on scenarios such as smart gri¬¬ds, wireless sensor networks (WSNs), and other IoT environments	Analysis of existing protocols, proposing a new protocol using PUFs, and conducting security and performance evaluations through simulations and theoretical analysis
[17] (2022)	Highlights the need for improved security against conventional and differential attacks on PUF-based key generation, and addresses the limitations in the existing lightweight two-factor authentication schemes for IoT devices	Uses PUFs combined with PKI digital certificates, along with cryptographic techniques like hashing and encryption	Include IoT environments such as smart homes, industrial IoT, and cloud-based IoT systems, focusing on secure user and device authentication	Detailed security analysis of existing protocols, the development of an improved PUF-PKI based authentication scheme, and validation using the Tamarin prover to ensure robustness against various security attacks

TABLE I
(CONTINUED)

Reference (Year)	Gaps Identified	Technologies Used	Applications Covered	Methodology
[18] (2021)	Highlights the need for secure communication in ITS, particularly in preventing data tampering and ensuring the integrity of vehicle accident detection and notification data	Utilizes blockchain technology combined with certificate-based authentication to ensure secure communication, data immutability, and transparency in vehicle accident detection and notification	Include ITS, focusing on secure V2V and V2I communication, specifically for vehicle accident detection and notification systems	Designing a new blockchain-enabled certificate-based authentication scheme, performing security analysis against potential attacks, and conducting performance evaluations through simulations and theoretical analysis
[19] (2022)	Highlights vulnerabilities in existing IoT environments and the lack of comprehensive security mechanisms for trust and authentication	The protocol uses trust aggregation, certificate-based authentication, ECC, and digital certificates	Focuses on edge-enabled IoT systems, including smart homes and industrial IoT settings, ensuring secure communication	Security analysis, performance comparison with existing schemes, and practical implementation using NS2 simulations to measure end-to-end delay, throughput, and packet loss rate

between different platforms in complex IoT environments. We also discuss the potential of using quantum computing to develop authentication protocols resistant to future technologies, such as quantum-based attacks, opening new avenues that have not been sufficiently explored before.

Thus, our research helps fill the gaps in the current literature by providing an integrated approach that combines security, efficiency, and sustainability in IoT authentication, enhancing the ability of connected systems to effectively and sustainably meet future security challenges.

D. Organization of the Paper

The outline of this paper is shown in Fig. 1. The remainder of this work is organized as follows. In Section II, we introduce the world of identity verification in Internet device authentication. We explore its different classifications and address its limitations. Moving forward, in Section III, we examine the latest trends in IoT authentication, focusing on machine learning and blockchain. In Section IV, we discuss the potential opportunities in this field and the role of advanced technologies in improving security.

The paper also includes an Insights Gained section, drawing key insights and guiding future research in Section V. It concludes by summarizing findings and making recommendations for future research in IoT authentication in Section VI.

This survey aims to provide a comprehensive classification of IoT authentication protocols, addressing the limitations identified in previous research and highlighting new trends and opportunities. This study offers an extensive overview of the current literature and emphasizes aspects that have not been sufficiently covered in previous studies.

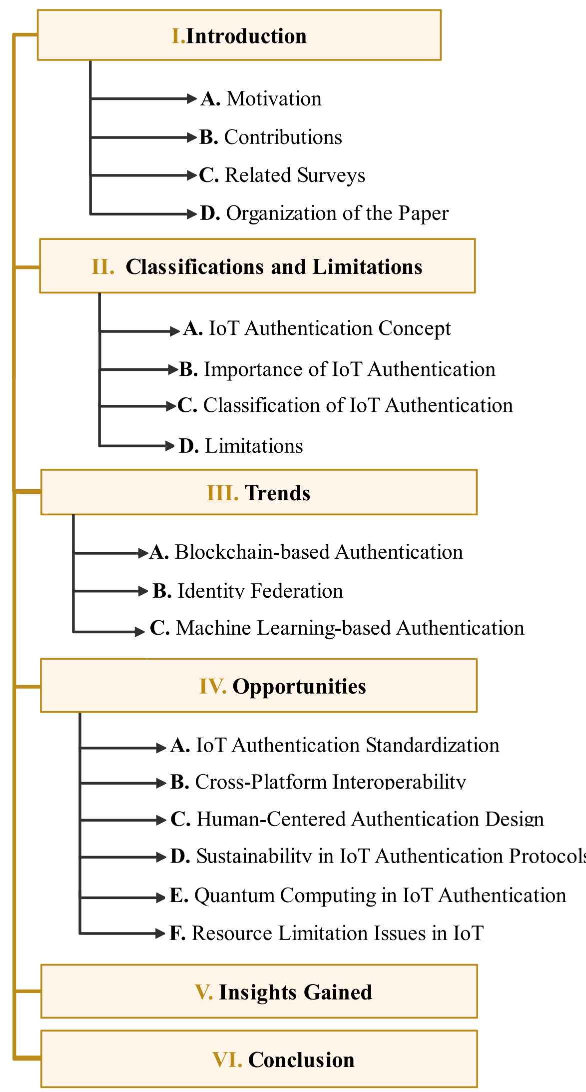

Fig. 1. The organization of the paper.

II. CLASSIFICATIONS AND LIMITATIONS

The concept of IoT authentication is provided in II-A; The importance of authentication is discussed in II-B; Classifications of IoT authentication protocols are provided in II-C, exploring how each class enhances security and trust in the world of IoT and exposes its advantages and challenges. Finally, at the end of this section, the limitations are provided in II-D and summarized in Table IV.

Our classifications include traditional authentication protocols, such as password-based and digital certificate-based methods, in addition to modern approaches like biometrics and multi-factor authentication. This comprehensive classification

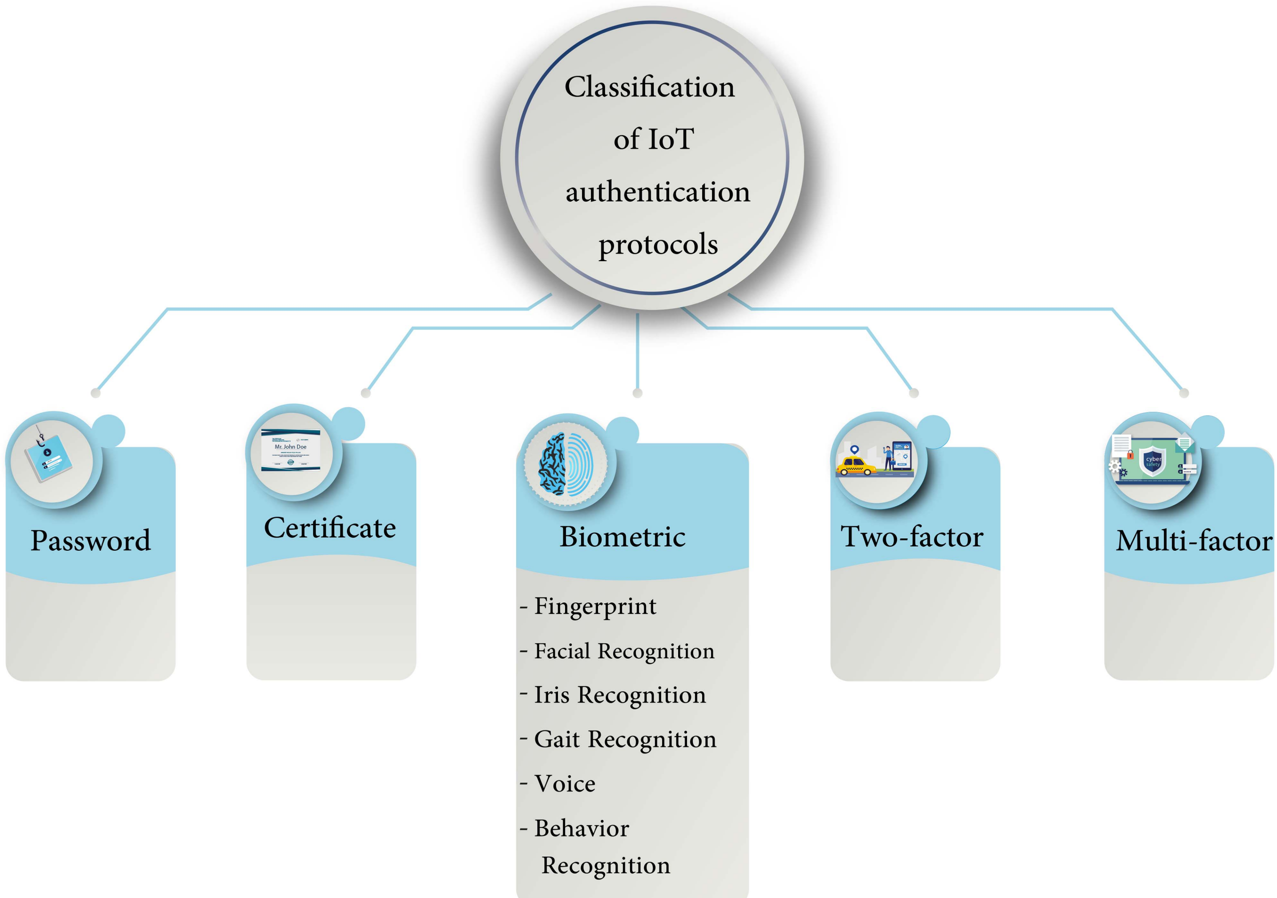

Fig. 2. Classification of IoT authentication protocols.

aims to provide a deeper understanding of how to improve security in IoT environments using various methods, distinguishing our study from previous research focused only on specific aspects.

A. *IoT Authentication Concept*

Authentication is a process that involves verifying the identity and integrity of devices and entities that participate in an IoT network. The purpose of authentication is to ensure that the participating devices are genuine and trustworthy and to provide secure communication and protection for transmitted and received data. This includes the use of secure protocols, encryption techniques, mutual authentication, key management, and digital signatures. IoT authentication involves both user authentication and device authentication. However, device authentication faces unique and significant challenges compared to user authentication. The authentication process is necessary to mitigate various types of attacks [1], [20].

When entering a website, you are often asked to go through a login process, typically using a username and password, in order to access and manage your posts under your name [3], [21]. We can explain the authentication process by answering the following question: How is it ensured that the user is authorized to write posts under their username and that they are indeed the same person as indicated by the login screen? In other words, what are the processes that the server goes through to verify the user's identity and authorization to publish under that name? The process begins with the client (the web browser) sending a request to the server, which is a POST request containing a set of information known as body parameters [1], [22]. These parameters include the data entered by the user, such as the username and password.

The server then queries the database to verify this user, looking for records that match the entered data. The database holds a table called the "users," which contains all user data. If a user with matching data is found, the server confirms the user's existence [22]. As a result, the server generates a token, a complex code such as 'Dre57iG39741fsBau', which is used to identify the user in subsequent requests, such as post-creation. The server then sends a request to the database to store this token in a table, associating it with the user's account. Additionally, the server includes this token in its response to the client (web browser), which then stores the token for future use in subsequent user requests [3]. Consequently, the browser redirects the user to the posts page, where they can create a new post. After the user enters the POST details and clicks the submit button, the client sends another post request to the server. This request includes

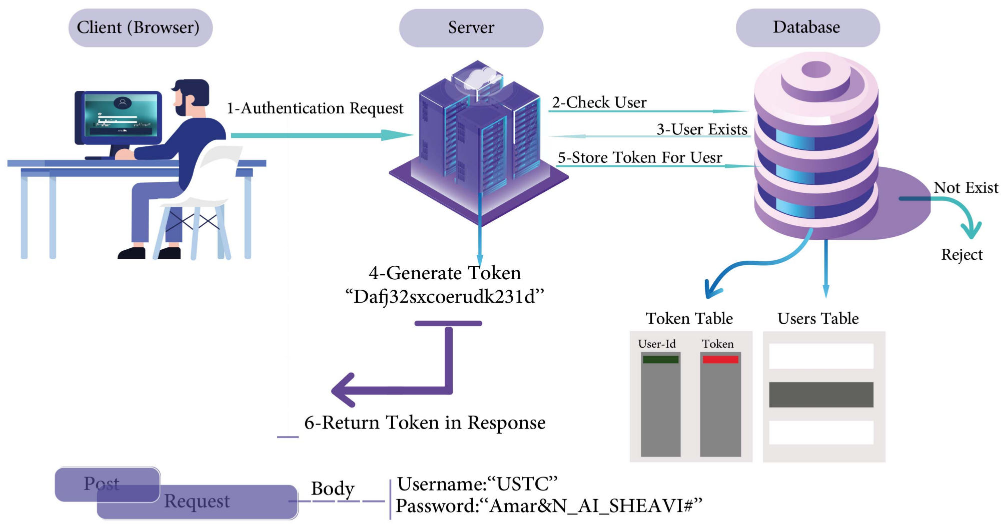

Fig. 3. Interactions between client, server, and database operations.

the body parameters specific to the post, such as title and details, and crucially, the token previously generated by the server and stored in the browser to identify this user. The server receives this token and queries the database to verify its existence and association with a specific user.

Once the database confirms the validity of the token, it sends a response to the server verifying the token and its association with the user [21]. During the same time, the server validates the post request and instructs the database to create a new post, storing the data as specified by the user. Subsequently, the server sends a response to the client confirming that the post has been successfully created [23]. Fig. 3 illustrates this process.

B. Importance of IoT Authentication

Effective authentication is critical to maintain the confidentiality, integrity and availability of IoT systems and data, as well as to prevent unauthorized access and attacks. In this research, we focus on understanding the fundamental pillars of IoT authentication and the security-related challenges. We assume that the importance of IoT authentication is based on driving factors such as the following perspectives: trust, identity verification, privacy, and protection [24]. These aspects will be discussed in detail below.

1) *Trust:* Authentication ensures that only authorized users or services have access to devices or data. For example, university systems that display student information assign each student a unique ID and password. It is essential that each student can only access his own data, ensuring trust by restricting access to authorized individuals [19], [25].
2) *Identity Verification:* To grant a user access to a device, his/her identity must be verified [27]. This can be achieved through various authentication methods [26], including:
   a) *Two-Factor Authentication (2FA):* Requires the user to provide two different forms of identification [33], such as a password and a verification code sent to phone [27], [34].
   b) *Biometric Identification:* Utilizes biometric features such as fingerprint or facial recognition to verify the user's identity [35]. These methods are considered highly secure and effective [36].
   c) *Shared Identity Verification:* This approach uses a shared identity among connected devices in the network [37]. IoT establishes a trusted shared identity [38].
   d) *AI-Enhanced Authentication:* Analyzes user behavior patterns using artificial intelligence and machine learning techniques to create a behavioral model [39]. This model is used to verify the user's identity and detect any unusual patterns indicating unauthorized access or attacks [40].
   e) *Location-Based Authentication:* Uses the user's geographical location information [41], comparing the actual location with the expected location based on data and context to verify his identity [28].
3) *Privacy and Device Security:* Each IoT device contains sensitive and important data that needs to be protected. Only authorized individuals should have access to this data [30]. Therefore, authentication ensures the privacy and security of devices by allowing access only to authorized users [29].
4) *Protection:* It is crucial to have policies in place to protect IoT devices from viruses, environmental disasters, and any potential threats or breaches [29], [31]. A good reputation is vital for the success of any system, company, or device. Protection is necessary from all angles [32]. Table II examines IoT security, emphasizing trust, identity, privacy, and protection. Notably, in the trust sector, blockchain systems are vital for ensuring data reliability in smart networks and online transactions [19], [24], [25], where the level of importance was classified based on insights drawn from referenced research. Table II illustrates the importance of different security aspects in IoT by categorizing them as "High" or "Very High" based on specific criteria such as

TABLE II
COMPARING IOT SECURITY ASPECTS: TRUST, IDENTITY, PRIVACY AND PROTECTION

Pillars of Security in IoT	**Scientific Function**	**Uses and Importance in IoT Security**	**Types of Devices**	**Real-life Applications**	**Importance**
Trust [24], [19], [25]	Establishing trust relationships	Building confidence in data or communication	Blockchain-based systems, digital certificates	E-commerce, online transactions, data sharing, IoT networks	High
Identity verification [26], [27], [28]	Authenticating user identity	Ensuring accurate identification	Biometric scanners, smart cards, facial recognition systems	Online-banking, access control, border control, identity verification	Very High
Privacy or device security [29], [30]	Protecting data and device	Safeguarding data and device from unauthorized access	Firewalls, encryption algorithms, antivirus software	Mobile devices, cloud computing, IoT devices, computer networks	High
Protection [31], [32]	Securing against threats	Shielding against malicious attacks	Intrusion detection systems, security cameras	Cybersecurity, physical security, network protection	Very High

impact level, application scope, and compensability. The classification follows the rule: High: if the security loss can lead to manageable issues that can be resolved quickly without long-term effects on the entire system, and Very High: if the security loss can cause severe and long-lasting problems resulting in significant losses or serious security damages that are difficult to address. Security aspects like "Trust" and "Privacy or Device Security" are categorized as "High" because they play crucial roles in building confidence and protecting data from unauthorized access. Their applications include e-commerce and cloud computing, where any security breaches can be quickly managed using additional measures like firewalls and encryption algorithms. On the other hand, "Identity Verification" and "Protection" are classified as "Very High" due to their critical roles in securing sensitive systems such as banking and border control, where security breaches can have severe consequences that are hard to mitigate. Identity verification ensures accurate user identification in critical applications, while protection defends against malicious attacks using intrusion detection systems and security cameras.

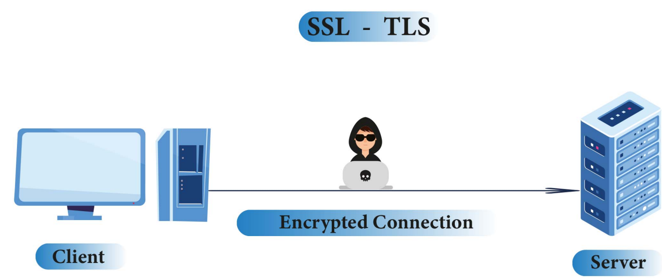

Fig. 4. Establishing an encrypted connection between the client and the server.

C. Classification of IoT Authentication

Authentication in IoT can be categorized into several types, including password-based, certificate-based, biometric, two-factor, and multi-factor authentication. Fig. 2 shows these classifications, and provides a clear visual representation of the different methods and their relationships.

1) Password-Based Authentication: Password-based authentication is common due to its simplicity and ease of use [42]. However, it faces security challenges and concerns, especially when using easily guessable or common passwords, making it susceptible to brute-force attacks and password cracking techniques [5]. This method verifies a user's identity in various applications across systems. Despite its widespread use, it has vulnerabilities, which prompts researchers to explore alternatives [1].

Researchers have focused on comparing password-based authentication methods, such as gesture-based and keystroke-based passwords. These methods aim to improve security and provide better resistance against motion-sensing attacks, in which unauthorized individuals attempt to discover the password [43]. Gesture-based passwords offer better resistance to attacks than traditional passwords but may raise usability concerns such as increased authentication time and error rates [44]. In contrast, keystroke-based passwords rely on distinctive typing patterns and have demonstrated their effectiveness in resisting motion-sensing attacks while providing better usability features [45].

Password-based authentication is widely used in IoT, from smart homes to healthcare. In smart homes, it is used to control applications such as lighting and security systems [46]. In smart cities, it helps with resource management systems, such as electricity distribution [14]. In smart healthcare, it gives access to electronic medical records and patient care systems. Despite their user-friendliness and cost-effectiveness, traditional passwords are vulnerable. Therefore, it is essential to adhere to best practices and implement additional security measures [1], [5].

2) Certificate-Based Authentication: Certificate-based authentication is a widely used secure mechanism to verify the identities of participants in digital communication, including users, devices, servers, and IoT devices [1]. Fig. 4 illustrates the authentication process for SSL-TLS security protocols used to establish an encrypted connection between the client and the server, securing the channel between the device and the web server via HTTPS. The core of this type of security lies in the use of digital certificates issued by trusted entities, containing relevant information about the entity's identity, public key, and other details [4]. The central concept of this mechanism revolves around a trusted third party known as a Certificate Authority (CA), which issues and signs digital certificates, as shown in

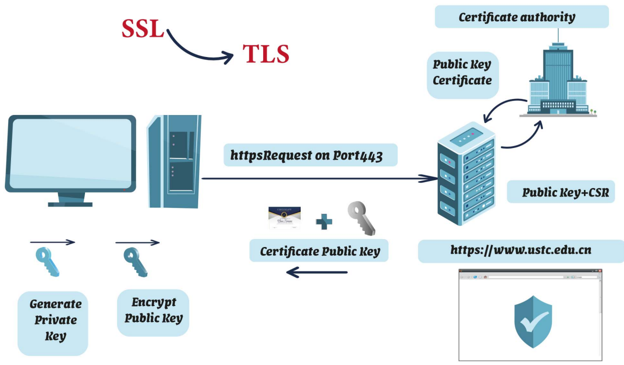

Fig. 5. The process of certificates and how it is done.

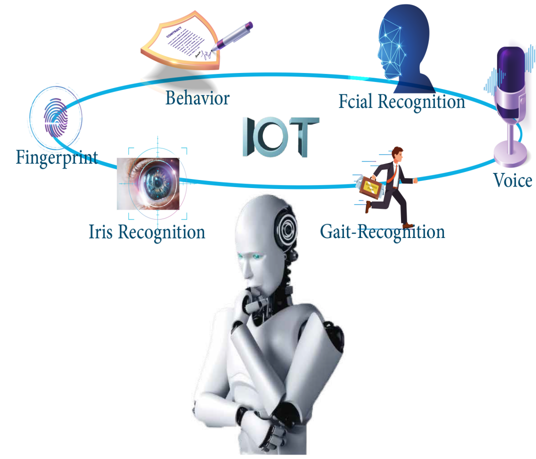

Fig. 6. Biometric authentication applications in IoT.

Fig. 5. These certificates serve as credentials that allow entities to securely and reliably prove their identity to other parties [1], [5].

Certificate-based authentication offers several benefits, including trust and reliability in digital communication, validation of certificates, and the trust authority issuing them, as well as solid identity confirmation [1], [17]. Secure authentication facilitates secure communication by establishing secure channels using the public keys included in the certificates [19].

Ensuring communication security between clients and network servers is crucial for data protection. The use of PKI verifies the identities of relevant parties, with CA playing a crucial role in proving its competency in the process by issuing a pair of encryption keys. The end entity retains its private key within a digital certificate linking the individual's name to their public key, ensuring the encryption and protection of sent and received data [5].

Certificate-based authentication provides numerous advantages, including the recognition and widespread acceptance of certificates in various applications and protocols, such as SSL/TLS, email, and cloud applications [17], [18]. The certificate-based authentication system can be extended to support a large number of users, devices, and applications [1]. Fig. 5 shows the authentication process and its implementation to achieve all the benefits of certificate-based authentication.

3) Biometric Authentication: Biometric authentication is based on the use of unique, measurable and verifiable characteristics in individuals to achieve high security and credibility in the authentication process [47]. This technology is based on the biological or behavioral characteristics of individuals, which are measured and analyzed to determine their identity. Biometric authentication has broad applications, as shown in Fig. 6, especially with the proliferation of IoT devices that can be worn [1], [5].

Physiological signals, such as electrocardiography (ECG), can be easily obtained without user intervention [48]. Distinctive factors for biometric authentication include the following:

a) *Fingerprint Authentication:* Fingerprint authentication is one of the most widely used features in biometrics. It involves measuring the ridges and furrows on the external surface of the finger to determine a person's identity [47]. The authentication process begins with a fingerprint reader device equipped with a sensor responsible for capturing a fingerprint image. The captured image is converted into digital data using techniques aimed at pattern recognition and the extraction of unique fingerprint features. These unique features are stored in the form of an encrypted fingerprint template that represents the biometric identity of the individual [49].
When a user wishes to authenticate, they simply place their fingerprint on the reader device. The unique features of the user's fingerprint are extracted from the current image and compared with the templates stored in the database. If there is a match between the current and stored features, the authentication process is successful and the user's identity is confirmed [50].
Fingerprint authentication is widely used in smartphones, portable computers, access control systems, and identity verification systems. It does not require expensive equipment and offers a high level of reliability and accuracy. Despite these advantages and conveniences, its performance can be weakened due to factors such as dry or damaged fingerprints. However, this limitation can be overcome by employing 3D fingerprint imaging in the authentication system [47], [49].

b) *Facial Recognition Authentication:* Face recognition authentication focuses on analyzing facial features such as the nose, facial wrinkles, eyes, and mouth to determine a person's identity.
The authentication process begins with a face recognition device that captures a user's face image using an integrated camera. The captured image is processed to extract unique facial features, such as the structure of the chin, the eyes, the mouth, and the nose [51]. These unique features are converted into digital data representing the biometric identity of the individual. When an authentication request is made, a new image of the user's face is taken to extract biometric features. The current features are compared to the stored user data to determine the degree of similarity. If the current features match the stored data sufficiently, the authentication process is completed, and the individual is accepted as a valid user [52].

Face recognition authentication provides a high level of security and convenience, as the face is a unique feature that easily distinguishes an individual from others and can be easily recognized. This method can be used for authentication in smartphones, security systems, and computers. Face recognition authentication excels in quickly verifying identity as it does not require direct contact [1], [53]. However, it may be less effective in low light environments or when significant facial feature changes are noticeable. This limitation can be overcome by employing 3D images or videos [52].

c) *Iris Recognition Authentication:* Iris recognition technology is used to assess the distinctive iris pattern to verify an individual's identity [54]. Authentication procedures commence with an iris reader device. The iris pattern is considered unique due to its distinctive tissues. The reader device employs specialized techniques in illumination and imaging to obtain an accurate iris image. Subsequently, the captured image undergoes processing to extract unique features, including the overall pattern and distinctive points of the iris [48].

These unique features are encrypted as digital data. When initiating an authentication request, the individual presents their eye to the reader device. The current iris features are then measured and compared to the stored data. If a match is detected between these features, the authentication process is completed, and access is granted to the individual [55].

It is worth noting that iris recognition authentication provides an advanced level of precision and security. Applications for this technology can be found in some government institutions, banks, and high-security access systems. It remains unaffected by changes in external appearance. However, its implementation is costly due to the need for specialized equipment, and the process may be complex or inconvenient for the user, requiring their cooperation to focus on the scanning device [56].

d) *Gait Recognition Authentication:* Gait recognition authentication relies on analyzing the unique walking pattern of an individual to verify their identity. It involves using sensors and computer analysis to measure and examine the unique gait pattern. Specialized sensors are used to measure gait parameters such as step length, height, and response time. These measurements are converted into digital data. During the authentication process, the user is asked to walk, and the current gait pattern is measured and compared to the stored data [57]. If a match is found, the process is completed successfully, and access is granted to the user.

This method is used in some high-security access systems and specific military applications. It is difficult to replicate or forge a gait pattern. However, it requires high recognition accuracy and appropriate imaging conditions. It is less effective in crowded areas or locations with fast movement [58].

e) *Voice Authentication:* Voice recognition authentication uses unique voice characteristics of an individual to verify their identity. The recorded voice is analyzed and compared to previously stored data for authentication purposes.

Biometric features derived from the user's voice are stored as digital information. During the authentication process, the user provides a passphrase or spoken word to extract biometric characteristics from their voice, which are then compared to the stored data. If a match is found between these features, the process is completed successfully and the user is granted access [59].

Voice recognition authentication provides a secure and convenient method, as each person's voice is considered a unique property used for identity verification. This type of authentication can be implemented in access control systems and smartphone services. Additionally, it can be used remotely without the need for physical contact with a specific device. However, voice recognition may be susceptible to noise and external disturbances and requires recording and storage of voice samples from the user [60].

f) *Behavior Recognition Authentication:* Behavior recognition authentication is a technology that verifies a person's identity by analyzing their behavioral patterns, such as typing or keystroke dynamics, and comparing them to stored data. This authentication method relies on the analysis of different behavioral patterns, such as typing or pressing keys on a keyboard, that are unique to each individual [60]. Specialized sensors, such as touch-sensitive keyboards or touch devices, capture these behaviors. The recorded data are then processed to extract the individual's unique behavioral characteristics, which are stored as an encrypted template representing their biometric identity. During authentication, the person's behavior is recorded, and the current features are compared to the stored data. If the characteristics match, authentication is accepted, confirming the user's trustworthiness [51].

Behavior recognition authentication provides a high level of verification and security because behavior patterns and actions are specific to each person. It finds applications in forgery detection and access control systems. It does not require expensive equipment or technical complexities, and the individual's behavior pattern can be continuously monitored. However, this authentication method relies on users' cooperation in recording their behavior patterns and is not well suited for instant or real-time recognition [35], [51].

We can observe that all these fundamental biometric authentication methods, such as fingerprint scanning, face recognition, iris scanning, gait recognition, voice recognition, and behavior recognition, are based on employing different techniques to capture unique biometric data. This data is then compared with stored information to establish secure authentication and validate the user's identity. Moreover, biometric authentication technologies require the collection and storage of sensitive data associated with the biometric characteristics of individuals. This raises privacy concerns and highlights the importance of securely managing these data. In addition, there may be legal and regulatory restrictions on biometric technologies in certain

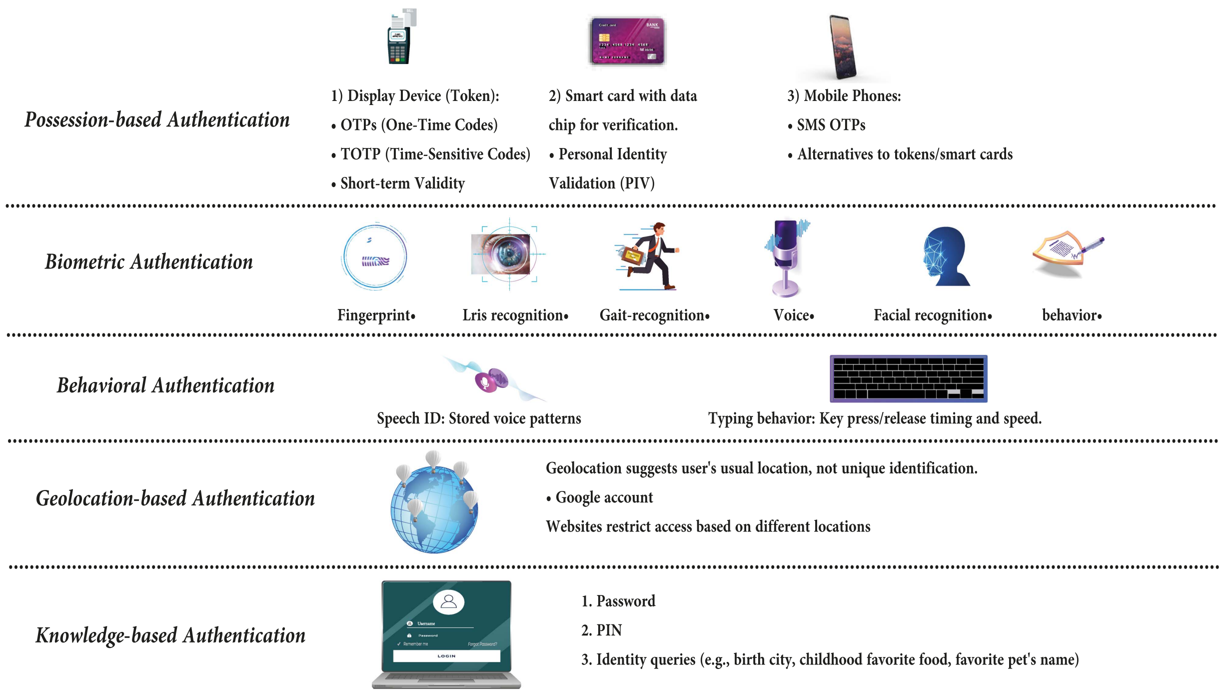

Fig. 7. Illustrates five common factors for multi-factor authentication.

countries, which include privacy rights and limitations on the use and collection of biometric data. These technologies may also face technical obstacles related to data analysis, system compatibility, and response time. However, despite these challenges, modern biometric authentication offers the most effective means of achieving credibility and security across various domains, including cloud computing, IoT devices, mobile devices, building access control, financial transactions, and identity verification in government systems.

4) Two-Factor Authentication: Two-factor authentication (2FA) is a security method that uses two factors to verify a user's identity: something known to the user, like a password (the first factor) and something possessed by the user, such as a one-time code, fingerprint or smart card (the second factor) [61]. These factors add an additional layer of security to IoT systems [62], [63].

In the 2FA process, a physical token, which connects to devices via USB, NFC, or Bluetooth, serves as the second factor. Users enter their password and then use the token for verification. Security tokens are secure, resistant to cyber threats, and offer portability with encrypted communication [64].

2FA improves security, reduces the risk of unauthorized access, and protects against password-related attacks such as brute force or password guessing [33], [65]. However, implementing these protocols can be challenging due to the diverse nature of IoT devices, resource limitations, and the need for precise design and integration, which may introduce inconvenience and user resistance [66], [67]. The challenges associated with 2FA include scalability, resource restrictions, and preservation of privacy [68], [69].

In summary, 2FA enhances security, particularly when using physical tokens, and safeguards against password-related attacks. However, its implementation requires addressing scalability, resource limitations, and privacy concerns, especially in IoT applications.

5) Multi-Factor Authentication: Multiple authentication methods, including password-based, certificate-based, biometric, two-factor, and multi-factor authentication, each have their strengths and limitations in IoT systems.

Multi-factor authentication (MFA) enhances security by requiring multiple pieces of information for identity verification [71], [72]. It relies on three key factors: something the user knows (like a password), something the user has (such as a mobile phone or smart card), and something the user is (biometric information) [73], [74]. Fig. 7 illustrates five common factors used in MFA [73].

MFA protocols strengthen security by integrating multiple authentication factors, making it difficult for attackers to impersonate authorized users [73]. They also reduce the risks associated with attacks that rely solely on stolen passwords or credentials [73], [75].

The design of MFA protocols involves accommodating various authentication factors and scalability for various IoT applications. However, implementing these protocols requires additional infrastructure, potentially increasing complexity and costs [66], [74]. Integrating these protocols with existing IoT systems may introduce additional steps, which could cause inconvenience to users [73], [76].

In summary, advancements in multi-factor authentication aim to heighten security and provide a seamless user experience, mitigating risks associated with breaches and fraudulent identity usage.

The Table III provided compares of a set of performance levels used to classify various authentication methods. These levels help to evaluate and compare these methods based on

TABLE III
COMPARISON OF AUTHENTICATION METHODS IN IOT: SECURITY, USABILITY, COST, SCALABILITY, CUSTOMIZATION, AND USAGE RATE

Authentication Method ⇒	**Password-based** ⇓	**Certificate-based** ⇓	**Biometric** ⇓	**Two-factor** ⇓	**Multi-factor** ⇓
Security level	Low to medium	High	High	High	Very high
Usability	High	Medium	Medium	Medium	Medium to low
Cost	Low	High	Medium	Medium	Medium to high
Scalability	Low	High	Medium	Medium	High
Customization	Medium	Medium	Limited	Medium	High
Usage rate	High	Medium	Increasing	Medium	Increasing
References	[5], [34]	[1], [17]	[70], [52]	[34], [66]	[17], [52]

specific criteria, particularly focusing on the aspects of security, usability, cost, scalability, customization, and usage rate. They are represented as follows:

(High): Indicates that the method or approach excels in this aspect compared to available alternatives. For example, if the security level is high, it means that the security is robust and highly dependable. (Low): Indicates that the method or approach exhibits poor performance in this aspect. For example, if the security level is low, the security is weak and may be vulnerable to attacks. (Low to Medium): Suggests that the approach falls in the lower half of the performance scale. (Medium): Signifies that performance is considered average without significant superiority or weakness. (Medium to Low): Indicates that the approach falls in the upper half of the performance scale. (Very high): Indicates that the performance in this aspect is excellent and significantly stronger than the alternatives. (Medium to Low): Suggests that the approach falls in the lower half of the performance scale. (Limited): Indicates that the performance in this aspect is limited and insufficient for some scenarios. (Increasing): Signifies that performance in this aspect improves over time with evolving technologies.

These levels were used to classify various authentication methods based on the perspectives of researchers and authors in the references mentioned. The purpose of employing these levels is to provide a concise and clear comparison between different methods and identify their anticipated advantages and disadvantages from the researchers' viewpoint.

These levels represent the outlook of the studies and references that were relied on in creating the table. These levels could be the conclusions of the researchers drawn from their analysis of the evidence available in the studies and experiments they conducted. The purpose is to guide the reader through the understanding of how researchers classify the level of security, usability, cost, flexibility, and usage rate for each type of authentication based on these studies.

It should be noted that these levels reflect the personal perspective of the researchers and their comparisons among these methods based on the information and results collected in the studies they conducted and presented in the references used.

D. Limitations

Table IV provides an assessment of various IoT authentication protocols, detailing methods, security measures, performance analyses, and their limitations. It illuminates the advanced technologies currently in use, particularly blockchain and lightweight authentication, which serve as key focal points for delivering exceptional security and efficient performance. However, the challenges surrounding IoT security are complex and multifaceted [77]. Devices do not function in isolation, but rather within an interactive ecosystem involving other devices and systems, thus introducing novel security complexities. Despite the ingenuity of these protocols, the primary challenge arises from the human factor, which is susceptible to targeting. This is where the role of user training and awareness becomes crucial [5], [8]. However, challenges persist in ensuring widespread adherence to security practices.

Specific data protection requirements dictated by laws and regulations in various industries contribute to the limitations. The rapid pace of technological advancement requires continuous assessment of protocol security and adaptation to modern technologies. Practical effectiveness remains a focal point, requiring ongoing efforts to address emerging security challenges effectively [8]. Although advancements in blockchain and lightweight authentication are evident, the limitations outlined here underscore the intricate nature of IoT security. Addressing these limitations is essential for establishing a secure and reliable foundation for adopting IoT technology.

In conclusion, the detailed examination of IoT authentication protocols highlights their critical role in securing interconnected devices and data. By categorizing these protocols ranging from traditional methods like password-based and digital certificates to modern approaches such as biometrics and multi-factor authentication, we can better understand their unique strengths and weaknesses. This section underscores the importance of authentication in establishing trust, verifying identities, ensuring privacy, and providing protection within IoT environments. Acknowledging the limitations and challenges of each protocol emphasizes the need for ongoing innovation and adaptation to keep pace with evolving security threats, thereby ensuring the robustness and reliability of IoT systems.

III. TRENDS

The use of blockchain technology is explored in III-A. In III-B, we discuss the importance of integrating identity federation for better authentication. Lastly, we will explore the role of machine learning in protecting smart devices in III-C.

This study addresses current trends in IoT authentication, including the integration of blockchain technology and machine

TABLE IV
ASSESSMENT OF IOT AUTHENTICATION PROTOCOLS: METHODS, SECURITY MEASURES AND ANALYSIS OF PERFORMANCE

Reference	Authentication protocols	Technique used	Goal	Security verification	Performance (+)	Limitation (–)
[78]	A lightweight authentication protocol for D2D-enabled IoT systems with privacy	Lightweight authentication	Privacy preservation	Assumed high	Efficient due to lightweight nature	Specific to D2D systems
[79]	A novel vector-space-based lightweight privacy-preserving RFID authentication protocol for IoT environment	Vector-space-based lightweight authentication	Privacy preservation	Assumed high	Efficient and scalable	Specific to RFID systems
[80]	A two-stage feature transformation-based fingerprint authentication system for privacy protection in IoT	Two-stage feature transformation	Privacy protection	Not mentioned	High accuracy	Biometric data challenges
[81]	Blockchain-assisted authenticated key agreement scheme for IoT-based healthcare system	Blockchain-assisted key agreement	Secure healthcare data	Blockchain integrity	Improved trust	Complexity due to blockchain
[82]	Data provenance for IoT with lightweight authentication and privacy preservation	Lightweight authentication with data provenance	Privacy preservation	Assumed high (BAN)	Efficient with data lineage	Specific use-cases
[83]	Efficient privacy-preserving authentication protocol using PUFs with blockchain smart contracts	PUFs with smart contracts	Privacy-preserving authentication	Blockchain integrity	Efficient due to PUFs	Smart contract overhead
[84]	Proposing secure and lightweight authentication scheme for IoT based E-health applications	Lightweight authentication	Secure E-health operations	Assumed high	Suitable for health devices	Specific to E-health
[85]	Secure and privacy-preserving RFID authentication scheme for internet of things applications	RFID authentication	Secure IoT operations	Not mentioned	Scalable	RFID specific
[1]	Modern authentication schemes in smartphones and IoT devices: An empirical survey	Empirical analysis	Understanding modern schemes	Not a protocol(Evaluation process for schemes)	Broad coverage	Not a specific scheme
[8]	A scalable protocol level approach to prevent machine learning attacks on physically unclonable function based authentication mechanisms for internet of medical	Prevent ML Attacks on PUF-based authentication	Secure medical devices	Assumed high	Scalability	Medical IoT specific
[86]	A security framework for IoT authentication and authorization based on blockchain technology	Blockchain-based authentication	Secure IoT operations	Blockchain integrity	Improved trust	Blockchain overhead
[87]	Access management of IoT devices using access control mechanism and decentralized authentication	Decentralized authentication	Secure access management	Assumed high (Inconclusive)	Decentralized benefits	Complexity due to decentralization
[88]	DAG blockchain-based lightweight authentication and authorization scheme for IoT devices	Lightweight with DAG blockchain	Efficient authentication	Blockchain integrity	Lightweight	DAG specific challenges
[89]	Efficient hybrid centralized and blockchain-based authentication architecture for heterogeneous IoT systems	Hybrid centralized and blockchain-based	Secure IoT operations	Assumed High	Hybrid benefits	Hybrid challenges
[46]	Insider attack protection: Lightweight password-based authentication techniques using ECC	Lightweight password-based authentication with ECC	Protect against insider attacks	Assumed high	Lightweight means efficient	Password-based challenges

learning, which offer improved security and predictive capabilities to identify potential threats in advance. Our study highlights the potential of these technologies and the current limitations they face, focusing on challenges such as data security, access control, and handling advanced attacks.

A. Blockchain-Based Authentication

The field of IoT is witnessing a constant evolution in authentication techniques, and one of the emerging trends that has caught considerable attention involves the verification of IoT devices through the use of blockchain technology. These techniques seek to achieve robust and effective authentication for a wide range of devices and interconnected entities. A prominent instance [90], introduces an efficient authentication approach employing blockchain technology, which results in lower authentication costs and increased security measures. Moreover, blockchain technology is exploited in [9] to achieve identity verification within multi-wireless sensor network systems. It integrates the unique characteristics of blockchain to enhance both the security and efficiency requirements of interconnected network entities. In addition, in [10], [91], the authors developed a blockchain-based group key agreement mechanism for interconnected devices, with the aim of improving security and efficiency. It demonstrates how blockchain can significantly improve IoT authentication, offering key insights to develop smarter, more secure applications in the future.

By examining current and previous studies within the field of IoT authentication protocols, it becomes apparent that there are compelling emerging patterns that warrant further investigation. Researchers can be guided toward creating comprehensive authentication protocols that harness the capabilities of blockchain technology to achieve increased levels of security and verification. Furthermore, the exploration of cutting-edge encryption techniques and artificial intelligence presents an avenue to fortify data confidentiality and identify potential security vulnerabilities. Additionally, it is worth contemplating the application of industrial artificial intelligence and biometric recognition in authenticating devices, while also enhancing the integration with biological identities. This approach extends beyond singular devices to encompass the achievement of authentication in expansive collectives and intricate settings. These trends can pave the way for a wide array of secure applications and also shape the future landscape of IoT.

B. Identity Federation

The integration of Identity Federation in IoT authentication protocols presents an important area for future research to establish a unified identity across various devices. Key insights from recent studies suggest the following:

- *The Vital Role of Identity Federation [92]:* Emphasizes the role of Identity Federation in securing communications among IoT devices.
- *Blockchain for Identity Integration [93]:* Investigates blockchain for identity integration across IoT and other domains.
- *Efficient Authentication via Identity Federation [94]:* Proposes simplified, secure authentication mechanisms through Identity Federation.
- *Addressing Challenges in Identity Federation [95]:* Highlights the inherent uncertainties in IoT authentication present challenges, necessitating innovative solutions.

By examining past and present research in IoT authentication protocols using Identity Federation, it is clear that there are significant areas for future exploration. The focus should be on developing comprehensive authentication protocols that integrate Identity Federation with cutting-edge technologies such as blockchain, advanced encryption, and artificial intelligence. Priorities include enhancing inter-domain communication security, providing adaptable identities for various applications, addressing uncertainties in authentication, and blending biological and technological methods for improved security. Deep learning and Artificial Intelligence (AI) can further refine authentication protocols. Through these approaches, researchers can deepen their understanding and devise innovative solutions to overcome challenges in the IoT authentication protocol domain.

C. Machine Learning-Based Authentication

Integrating machine learning into IoT authentication can change the security field with new innovative methods such as pattern recognition and efficient models, driving advancements in smart device protection.

- *Authentication Methodologies:* The potential of integrating machine learning to perform sophisticated authentication methodologies within smart devices and the IoT is explored in [1], including the application of pattern recognition techniques and the utilization of biometric data.
- *Lightweight ML Models:* A structural framework designed for authenticating intelligent IoT devices by leveraging machine learning is introduced in [96], aiming at seamless and secure authentication process through the utilization of lightweight machine-learning models.
- *Categorization:* In [97], the authors presented a categorization, identifying obstacles, and furnishing guidance for forthcoming research trajectories in the domain of employing machine learning for authentication and authorization within IoT.

More research is needed to develop ML and AI models to identify security vulnerabilities, improve authentication accuracy through ML and biotechnologies, and design protocols immune to ML-based attacks. This is expanding to address privacy concerns and adapt to new threats, to enhance security knowledge and innovation.

In summary, examining current trends in IoT authentication discovers key advancements in security enhancement, blockchain technology offers secure identity verification and group key agreements, improving overall security measures. Identity federation supports unified identity management across devices, enhancing communication security and solving authentication challenges. Machine learning introduces advanced methods like pattern recognition and lightweight models, driving innovations in smart device protection. These trends highlight the need for continuous research to address emerging security threats and improve the reliability of IoT authentication systems.

IV. Opportunities

In IV-A, we explore the use of a common authentication framework to ensure the secure and reliable exchange of data between these devices. In IV-B, we discuss the potential of cross-platform interoperability. Furthermore, we emphasize the development of a human-centered authentication design in IV-C. In IV-E, we discuss quantum computing in IoT authentication. Finally, in IV-F, we address the issues of resource limitation in IoT.

This study identifies future research areas, including exploring the impacts of quantum computing on IoT authentication and developing resource-efficient protocols to overcome the inherent constraints of IoT. We recommend leveraging technologies such as machine learning, blockchain, and artificial intelligence to achieve adaptive authentication and privacy preservation techniques. In the domain of smart devices, certain limitations impact the seamless integration of authentication protocols. Challenges persist despite the emphasis on cross-platform harmony, human-centered design, and the implementation of authentication norms. Creating seamless interaction across diverse devices for an enhanced user experience and overcoming hurdles related to technology disparities and interoperability issues. Although facial recognition and fingerprinting methods offer a balance between convenience and security, they are not without limitations. Factors such as environmental conditions and device quality can affect the accuracy and reliability of these methods. The call for unified authentication standards to aid in data exchange and sustainability acknowledges an existing limitation. Establishing and implementing such standards on a global scale is a complex task given the diversity of devices, protocols, and regulatory environments. This complexity presents challenges in ensuring consistent and standardized authentication practices.

A. IoT Authentication Standardization

Smart devices play a crucial role in our everyday lives, offering a wide range of applications in sectors such as healthcare and industry, making them indispensable in our contemporary technological landscape. However, sharing data among these devices introduces increasing security and privacy complexities. The inclusion of illumination on their applications and obstacles increases understanding of the importance of embracing uniform authentication standards within the IoT framework [98].

Focusing on the authentication quandary reinforces the safeguarding of data transmission between devices. Leaning on unified benchmarks can be a transformative remedy to the increasing security predicaments. A comprehensive report underscores the gravity of crafting these standards to achieve secure and reliable data interchange and to ensure seamless communication across various devices [98], [99].

The rapid evolution of IoT technologies requires prioritizing human-centric authentication design. Involving users in the development process can increase ease of use and security. Technological ventures such as facial recognition, fingerprint and voice recognition could cater to user needs, providing a smooth and convenient authentication experience [80], [98]. In essence, adopting a standardized authentication framework for smart devices constitutes a fundamental step toward securing effective communication among diverse devices within the IoT ecosystem. By harmonizing security, user-friendliness, and addressing user needs, authentication technologies capable of tackling present and future challenges can be realized [88], [98].

B. Cross-Platform Interoperability

The domain of intelligent entities has become an inseparable aspect of our lives, where many platforms and devices collaborate within this interconnected framework. However, the challenge of achieving harmony across platforms presents itself as a pivotal determinant influencing user engagement and overall efficiency. It is important to note that this challenge originates from the disparity in protocols and configurations between these diverse platforms [99]. Failure to tackle this challenge could result in a decline in user satisfaction and impede the intended outcomes of intelligent technology. Furthermore, the lack of harmonization could increase the expenses associated with the development and integration of these disparate platforms [1], [99]. Consequently, the importance of formulating unified criteria and protocols to boost cross-platform compatibility becomes apparent. These criteria play a role in facilitating the seamless exchange of data and information among these heterogeneous platforms, thus enhancing the user experience and fully leveraging the advantages of intelligent technology [5], [99]. In addition, the achievement of cross-platform interoperability has the potential to broaden the horizon of intelligent technology applications across various domains, including industry, healthcare, and environmental contexts. This opens the windows to more sophisticated and efficient applications that contribute to the improvement of both quality of life and professional endeavors [99].

C. Human-Centered Authentication Design

In IoT technology, consistently highlighting the pivotal role of the human user is imperative. Smart technology requires a design that prioritizes human aspects, embodying the authentic essence of technology as a tool to amplify and serve human life [68], [80]. When designing human-centric authentication systems, finding the perfect balance between system security and user experience becomes very important. Authentication processes should be straightforward and user-friendly, ensuring a smooth experience for users. Currently, these processes can be characterized as high reliability to face security challenges [5], [80]. Instances such as facial recognition and fingerprinting techniques are examples of human-centric authentication. These technologies combine convenience and security, ensuring seamless user experiences while maintaining a commendable level of safety. However, the focus should revolve around maintaining an equilibrium between precision and privacy, avoiding any compromise on user experience [1], [80]. Practically speaking, a human-centered approach to authentication design requires actively involving users in the development process. This could be supported through surveys and user interactions, showing their requirements and expectations. This approach helps to develop technologies that cater satisfactorily and effectively to these needs [5], [80].

D. Sustainability in IoT Authentication Protocols

As our daily reliance on smart devices continues to grow, the importance of developing authentication protocols that focus on sustainability also increases, ensuring these systems remain efficient and secure over the long term is not just a technical challenge but also an environmental necessity.

- *Enhancing Energy Efficiency in Authentication Protocols:* Reducing energy consumption is a key objective in making authentication protocols more sustainable, techniques like Chaotic Maps provide innovative solutions for developing lightweight authentication protocols that consume less energy. These techniques help minimize the computational processes required, significantly lowering the energy consumed during repeated authentication processes for connected devices. Implementing such protocols enhances energy efficiency in IoT systems and reduces the environmental impact by decreasing the demand for energy, thereby easing the burden on both renewable and non-renewable energy sources [100], [101].
- *Reducing Carbon Footprint through Innovations in Authentication:* Innovations in authentication technologies offer a unique opportunity to reduce the carbon footprint of IoT systems, lightweight authentication protocols reduce the need for large data centers, which in turn lowers energy consumption and carbon emissions. Moreover, incorporating technologies like artificial intelligence to manage and improve protocol efficiency can lead to higher performance with less energy consumption, further minimizing negative environmental impacts [102], [103].
- *Using Blockchain Technologies to Enhance Sustainability:* Blockchain technologies play a fundamental role in developing sustainable and efficient authentication protocols, these technologies enable the creation of decentralized authentication systems that reduce resource consumption and improve operational efficiency. By distributing data across a network of nodes, the need for intensive computational power required by traditional systems is reduced,

leading to lower energy and resource consumption. This shift towards adopting blockchain technologies not only enhances sustainability, but also creates a safer and more energy-efficient environment, making it an ideal solution to the current and future environmental challenges faced by IoT systems [104], [105].

Embracing sustainable practices in authentication protocols is a vital step toward building more efficient and sustainable IoT systems, by developing protocols that focus on reducing energy consumption and carbon emissions, and by adopting blockchain technologies, we can enhance the capabilities of current and future systems to meet security and sustainability requirements in an integrated manner. This approach not only responds to environmental challenges but also represents an opportunity to achieve sustainable technological progress that meets the needs of future generations.

E. Quantum Computing in IoT Authentication

Quantum computing represents a significant advancement in IoT security by offering quantum-resistant algorithms capable of addressing advanced security threats. Techniques such as certificateless authentication and blockchain-based frameworks resistant to quantum attacks are powerful tools to improve security and prevent unauthorized access [106]. Quantum encryption algorithms can secure data and verify device identities in a tamper-proof manner [107].

Additionally, quantum-based authentication and the integration of blockchain and quantum computing provide innovative solutions for authentication challenges in resource-constrained IoT environments. These technologies contribute to high security levels, boosting trust in connected systems and improving the efficiency and sustainability of IoT technology [108]. By leveraging these innovations, we can build a more secure and reliable digital infrastructure that aligns with future digital aspirations [109]. Recent studies demonstrate that quantum technologies can effectively protect against complex cyberattacks, opening new possibilities for secure IoT applications [110], [111]. In the context of our research, these advancements highlight the importance of developing advanced authentication protocols capable of addressing future security challenges, thereby enhancing the security and reliability of IoT.

F. Resource Limitation Issues in IoT

One significant opportunity in the field of IoT authentication lies in addressing the resource limitation issues that characterize IoT devices. These devices often operate with limited resources such as processing power, memory, and battery life [112], which pose challenges to implementing security measures. To overcome these challenges, lightweight authentication protocols that require minimal computational resources are being developed [113]. Additionally, machine learning and artificial intelligence techniques are being used to create adaptive authentication systems that intelligently manage resource allocation [112], ensuring efficient and secure operations and enhancing the security and reliability of IoT systems.

Overall, standardizing authentication frameworks is essential for ensuring secure and reliable data exchange among diverse devices, enhancing communication, and addressing security challenges. While cross-platform interoperability is vital for seamless integration and user satisfaction, reducing costs and improving the effectiveness of intelligent technology applications across various domains. Human-centered authentication design emphasizes the importance of user-friendly, secure systems that balance convenience with high security, ensuring a positive user experience. These opportunities highlight the need for ongoing innovation to overcome existing limitations and enhance the overall security and efficiency of IoT systems.

V. Insights Gained

Through our comprehensive analysis of the importance of authentication and its classifications in the world of connected things, our exploration of future horizons and promising opportunities in IoT technologies, as well as our understanding of the constraints and complexities in their authentication systems, we uncover valuable insights and guide research towards new pathways in the world of connected things. The insightful observations we have gained are as follows:

Machine Learning and AI-Based Authentication: We highly recommend exploring ML and AI for adaptive authentication in dynamic IoT environments, detecting anomalies, and identifying unauthorized access.

Zero-Trust Security Models: Proposing a shift to zero-trust models in IoT, requiring continuous authentication for each communication, aiming to enhance security and prevent lateral movement.

Multi-Factor Authentication (MFA): Developing tailored MFA techniques for IoT, combining factors like unique identifiers, passwords, and biometric traits for stronger security.

Blockchain-Based Authentication: Investigating decentralized, tamper-resistant blockchain technology for authentication in IoT to establish trust, ensure data integrity, and prevent unauthorized access.

Usability and User Experience: Emphasizing user-centric design, intuitive interfaces, and seamless integration of authentication processes into IoT devices to enhance usability and user acceptance.

Secure Device Onboarding: Highlighting the importance of secure mechanisms for device registration, provisioning, and initial authentication to establish trust in IoT networks.

Privacy-Preserving Authentication: Addressing privacy concerns by exploring protocols that minimize personally identifiable information, ensure user anonymity, and protect sensitive data while maintaining secure IoT authentication.

Sustained Collaboration among researchers and software developers.

In summary, the exploration of machine learning and AI for adaptive authentication, the adoption of zero-trust security models, and the development of multi-factor authentication techniques tailored for IoT are essential for enhancing security. Investigating blockchain-based authentication offers a decentralized and tamper-resistant approach, while focusing on usability ensures user-centric design. Additionally, secure device onboarding and privacy-preserving authentication protocols are crucial for establishing trust and protecting sensitive data. Sustained collaboration among researchers and developers is necessary to drive innovation and improve the security and performance of IoT systems. These insights pave the way for creating a safer and more efficient IoT environment.

VI. Conclusion

This paper explores the various aspects of authentication systems in IoT, highlighting the significant contribution of various classifications in improving security. New technologies such as machine learning and blockchain play a key role in improving authentication systems, such as improving the security of connected medical devices and strengthening networks in smart industries. Depending only on traditional technologies is not enough to deal with the intricate security risks present in IoT environments. This underlines the need for flexible and advanced security strategies, such as using blockchains for identity management across medical device networks, employing machine learning for data analysis, and predicting security threats in smart industrial infrastructures. However, several challenges need to be addressed when developing or implementing authentication systems for IoT. These challenges include privacy concerns, technical compatibility issues, and response time. The need for unified standards and user-focused designs, essential for improving usability and security, becomes apparent. We conclude that biometric technologies provide effective and secure identity verification methods, but come with their own set of challenges, such as the storage of sensitive data and privacy concerns. Overall, this paper makes a valuable contribution to broadening our understanding of authentication systems in the IoT, paving the way for new explorations that could shape the future of security and privacy in this evolving field.

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